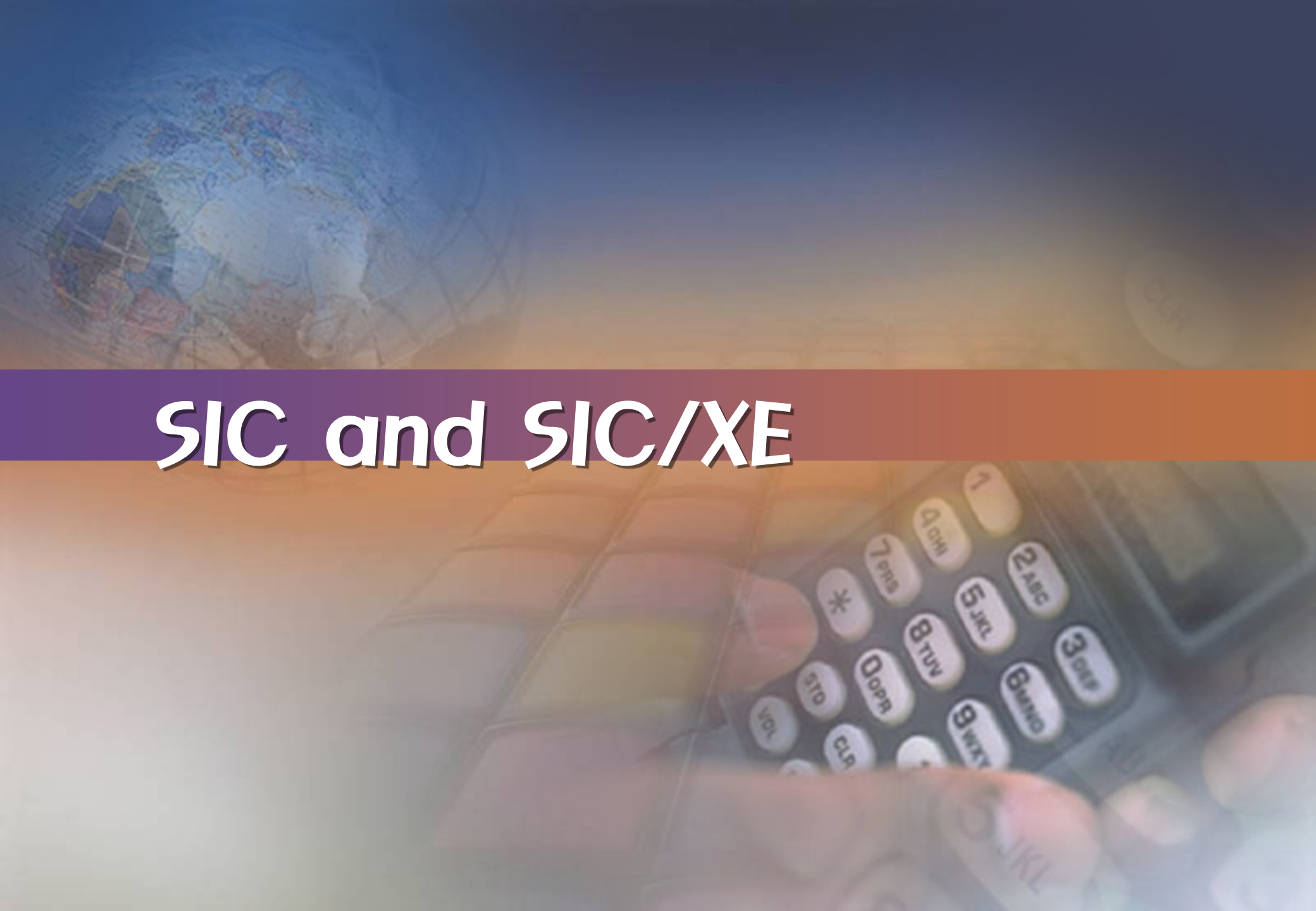


The background of the slide is a composite image. The upper left portion shows a stylized, textured globe with blue and white tones. The lower right portion shows a close-up of a hand using a calculator, with the keypad and display visible. A horizontal band of purple and orange gradient separates the globe from the calculator.

SIC and SIC/XE

수업 목표

- SIC/XE(Simplified Instructional Computer/Extra Expensive)의 개념을 이해한다.
- SIC/XE 명령어 구조와 사용방법에 대해 학습한다.
- 시스템 소프트웨어와 컴퓨터 시스템과의 관계를 이해한다.

SIC/XE Machine Architecture

- Max memory size : 1 Mbyte (2^{20} bytes)
- Registers (Additional)

Mnemonic	Number	Special Use
B	3	Base register; used for addressing (24 bits)
S	4	General working register (24 bits)
T	5	General working register (24 bits)
F	6	Floating-point accumulator (48 bits)

S, T register is only used
for storing data

Data Formats (SIC/XE)



- **48-bit floating-point data type**

- : sign of fraction (0=positive, 1=negative)

- : the high-order bit of the fraction must be 1

- $-0 \leq \text{fraction} < 1$

- : unsigned binary number (0 ~ 2047)

- Absolute value = fraction * $2^{(\text{exp} - 1024)}$

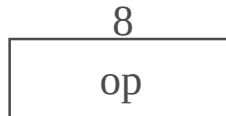
- Value of zero: setting all bits to 0

Instruction Formats (SIC/XE – I)

- How to support 1 Mbyte addressing
 - 2 possible options
 - Use relative addressing
 - Extend the address field to 20 bits

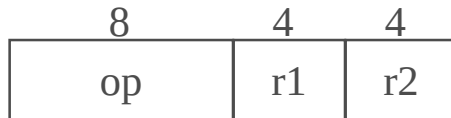
Instruction Formats (SIC/XE - II)

Format 1 (1 byte)

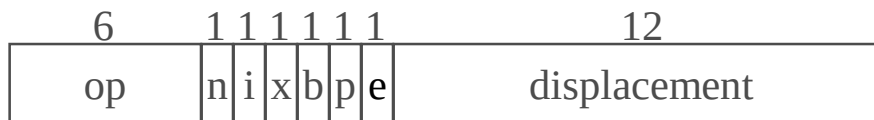


1 형식과 2 형식 명령어는 메모리 접근이 필요 없는 명령형식이다.

Format 2 (2 byte)



Format 3 (3 byte)



각 플래그 비트는 주소지정방식과 명령어 형식을 구분하기 위해 사용된다.

예) 플래그 비트 $e = 0$, 3 형식 명령어, $e=1$, 4 형식 명령어를 구분

Format 4 (4 byte)



이 동영상 교과목 자료는
GJ-RIP 사업의
지자체-대학 협력기반 지역혁신 사업비로 개발되었음.